



TECHNOLOGICAL FUNDAMENTALS IN THE BIG DATA WORLD

DISTRIBUTED PROTEIN MATCHING IN PYTHON

- To create a program to match a pattern introduced using the keyboard against all the proteins in the file provides using Python.
 - ▣ File: proteins.csv

- You are asked to:
 - ▣ Write the serial version of the program
 - ▣ Write the parallel version of the program using MPI
 - ▣ Write a report explaining your results (maximum 12 pages)

□ Make a program in Python that:

1. Reads the pattern to search from the keyboard.
2. Changes the pattern to UPPERCASE
3. Start metering exec time
4. Reads the protein patterns from the file.
5. Look for occurrences of the pattern string inside each protein sequence
6. If there are occurrences, register the id of the protein and the number of occurrences in the sequence
7. Print a barchart using protein id as X and number of occurrences as Y (you can use matplotlib.pyplot). Show 10 proteins with more matches. In case of coincidence, those with maximum hydrofob.
8. Print the protein id with max occurrences. If there are several, print the one with maximum hydrofob.

□

□ Run the command:

```
$> python protein-generator.py row_no <seed>
```

For development:

```
row_no 500000 <seed>
```

For testing and delivery (minimum):

```
row_no 5000000 <seed>
```

- Multiprocessing version with MPI
 1. Write a parallel version of you program using MPI
 2. Measure the time and optimize the program to get the fastest version you can.

- ❑ Make the lab in groups. Maximum 4 students.
- ❑ To deliver:
 1. Authors.txt file
 2. Python programs with all versions
 3. Written report
- ❑ Program names:
 - ▣ serial-proteins.py
 - ▣ mpi-proteins.py
- ❑ Deadline: **October 13th. 2026. 23:30 hours.**